

ARASH AHMADI

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EDUCATION

- **Doctor of Philosophy (Ph.D.), Electrical and Computer Engineering** Aug 2024 – Current
University of Oklahoma, Norman, OK, USA
Research focus: Fine-tuning small language models for efficient deployment on resource-constrained edge devices, with applications in agentic systems and diverse domains.
- **Master of Science (M.S.), Electrical and Computer Engineering** Aug 2024 – Spring 2026
University of Oklahoma, Norman, OK, USA
GPA: 3.88/4.0
Thesis: *Search-Based Reward Function Optimization for LLM Reasoning via Reinforcement Learning*
- **Bachelor of Computer Engineering, Software Engineering** Sep 2018 – Feb 2023
University of Kurdistan, Sanandaj, Kurdistan, Iran
GPA: 3.88/4.0
Thesis: *Efficient brute-force state space search for Yin-Yang puzzle*

PUBLICATIONS

Journal Articles

- [J1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "Improving Aviation Safety Analysis: Automated HFACS Classification Using Reinforcement Learning with Group Relative Policy Optimization," *Expert Systems with Applications*, Article 132963, 2026. doi:10.1016/j.eswa.2026.132963.
- [J2] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "A comparative study of sampling methods with cross-validation in the fedhome framework," *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 3, pp. 570-579, 2025. doi:10.1109/TPDS.2025.3526238.
- [J3] **A. Ahmadi** and A. Khorramian, "Efficient Brute-force state space search for Yin-Yang puzzle," *The Journal of Supercomputing*, vol. 80, no. 3, pp. 3066-3088, 2024. doi:10.1007/s11227-023-05565-w.

Conference Proceedings

- [C1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "LLM-Powered HFACS Analysis of ASRS Narratives via MCP Bridge for Enhanced Aviation Safety Debriefs," in *AIAA SCITECH 2026 Forum*, p. 1195, Jan 2026. doi:10.2514/6.2026-1195.
- [C2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "Towards Transparent Artificial Intelligence: Exploring Modern Approaches and Future Directions," in *2024 Conference on AI, Science, Engineering, and Technology (AIxSET)*, pp. 248-251, IEEE, 2024. doi:10.1109/AIxSET62544.2024.00047.

Preprints & Under Review

- [P1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "Enhanced LLM Reasoning by Optimizing Reward Functions with Search-Driven Reinforcement Learning," *arXiv preprint arXiv:2605.02073*, 2026. doi:10.48550/arXiv.2605.02073.
- [P2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "MCP Bridge: A Lightweight, LLM-Agnostic RESTful Proxy for Model Context Protocol Servers," *arXiv preprint arXiv:2504.08999*, 2025. doi:10.48550/arXiv.2504.08999.

INTELLECTUAL PROPERTY

- **A. Ahmadi**, Y. M. Banad, and S. Sharif. "Artificial Intelligence-Powered Personal Computer Management System and Methods," U.S. Provisional Patent Application No. 63/736,972, filed December 20, 2024.

RESEARCH EXPERIENCE

- **Graduate Research Assistant, Inquire Lab** Aug 2024 – Present
University of Oklahoma, Norman, OK, USA (Supervisors: Dr. Yaser Mike Banad, Dr. Sarah Sharif)
 - My research focuses on optimizing and fine-tuning language models for deployment on resource-constrained edge devices. This work explores their applications in agentic systems and diverse domains such as aviation safety and health monitoring.
- **Research Assistant** Feb 2023 – Aug 2024
University of Kurdistan (Supervisor: Dr. Amanj Khorramian)
 - Conducted research on theoretical computer science projects in the fields of Graphs, Puzzles, and Parallel Algorithms.
 - Proposed, implemented, and parallelized a novel algorithm for the Yin-Yang puzzle; this work included analyzing the results and authoring the majority of the publication.

AWARDS AND HONORS

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- William H. Barkow Graduate Fellowship, University of Oklahoma 2025-2026
 - William H. Barkow Scholarship, University of Oklahoma 2025-2026
 - Journal Paper Award, Electrical and Computer Engineering, University of Oklahoma 2025
For the publication: "A comparative study of sampling methods with cross-validation in the fedhome framework"

TEACHING AND MENTORING

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- **Instructor and Lead Developer** 2025, 2026
Cybersecurity Essentials Workshop, University of Oklahoma
 - Led a 10-person team in developing and delivering a five-day intensive workshop for IT professionals, sponsored by the Oklahoma Office of Homeland Security, Oklahoma Homeland Security, and FEMA.
 - Designed the 2025 workshop agenda and description, shaping the instructional flow, topics, and participant-facing materials.
 - Authored and presented modules on emerging cyber threats, ransomware prevention, phishing analysis, secure network design, and firewall best practices.
 - Designed and developed the official workshop website and a custom content management platform.
 - **Teaching Assistant** Fall 2019 – Spring 2020
University of Kurdistan
 - Courses: Advanced Programming with Java (Spring 2020), Fundamentals of Programming in C (Fall 2019).
 - Conducted weekly review sessions and provided direct feedback on projects and assignments.

PROFESSIONAL AND COMMUNITY SERVICE

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- **Ad hoc Reviewer**, *IEEE Transactions on Artificial Intelligence* 2026
 - **Volunteer**, The Big Event, University of Oklahoma 2026
 - **Volunteer**, The Little Event, University of Oklahoma Oct 2026

SELECTED TECHNICAL PROJECTS

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- **Inquire Lab Research Website (inquirelab.ai)**: Designed and developed the official website for the Inquire Lab with Next.JS. The site serves as a public-facing platform to showcase research projects, publications, and team members.
 - **Traffic Prediction in Edge Environments (Python, 2021)**: Implemented a federated learning approach with a GRU network in TensorFlow to predict traffic. This method enhanced privacy and reduced communication costs.
 - **Task Scheduling Optimization (Python, 2020)**: Developed an application with a UI to implement and compare heuristic search algorithms (e.g., Greedy, Hill-climbing, Genetic Algorithm) for task scheduling.

LEADERSHIP AND ACTIVITIES

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- **Founder**, *LLM Engineering Club*, University of Oklahoma Mar 2026 – Present
 - Founded and organized a registered student organization for hands-on LLM engineering, responsible AI practice, workshops, study groups, and collaborative AI projects.
 - Organized club programming around lab-approved access to a computing cluster for local model training, fine-tuning, and evaluation.
 - Led AI news discussions and designed a workshop series where students train a small language model to play a Codenames-style word-reasoning game.
 - **Engineering Officer**, Game Developer's Association, University of Oklahoma Aug 2024 – Dec 2024
 - Delivered technical presentations on engineering principles in video game development.
 - Developed an interactive Ouija board powered by the Gemini 2.0 Flash large language model for a university-wide Halloween escape room event.

REFERENCES

Yaser (Mike) Banad, PhD

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Sarah Sharif, PhD

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School of Electrical and Computer Engineering
The University of Oklahoma
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